

Varun Shamanna

Bioinformatician & Genomic Epidemiologist

To work pragmatically in an organisation & pursue my research in a productive and challenging environment that provides me with the opportunity to improve my knowledge and enhance my skills

EXPERIENCE

Elucidata Inc, Bengaluru — Bioinformatics Engineer II

Feb 2026 - Current

- Single and Bulk RNA seq analysis for cancer and inborn metabolic disorders
- WGS analysis to call and annotate somatic variants between the normal and tumour samples
- Building custom AI agents to automate workflows

Microbial Informatics & Genomics Group, Central Research Laboratory, KIMS, Bengaluru — Senior Bioinformatician

July 2019 - Jan 2026

- Leading a team of Bioinformaticians to drive microbial informatics.
- We analysed more than 6000 whole-genome sequences of WHO ESKAPE pathogens.
- Created nextflow pipelines to analyse WGS data from routine sequencing.
- Developing scripts using Bash, R & Python to analyse the species-specific datasets.
- Metagenome analysis of Endometrium samples (EMMA test)
- Server and IT administration.
- Experience with sequencing on the Miseq & Nanopore platforms for substudies.

EDUCATION

Visvesvaraya Technological University (VTU) - PhD

Dec 2021 - Current

Microbial Genomics.

RV College of Engineering, Bengaluru — M.Tech

SEP 2017 - JUN 2019

Bioinformatics.

NMAM Institute of Technology, Udupi — B.E

AUG 2013 - MAY 2019

Biotechnology

Bengaluru, 560085

+91 8123341361

varunshamanna4@gmail.com

[Linkedin](#), [Twitter](#), [GitHub](#)

SKILLS

Next Generation Sequencing (NGS):

- Single Cell RNA seq analysis
- Human WGS variant analysis- Somatic/Germline
- Whole Genome Sequencing using Illumina MiSeq and Oxford Nanopore platforms
- Nanopore basecalling, demultiplexing, assembly, polishing, and variant calling

Bioinformatics & Genomic Data Analysis

- WGS analysis: QC, assembly, AMR prediction, serotyping, SNP analysis, phylogenomics, plasmid reconstruction, genomic epidemiology
- Bulk RNA-seq analysis: QC, alignment/quantification, DESeq2 differential gene expression, pathway analysis
- Single-cell RNA-seq analysis: Cell Ranger preprocessing, Seurat workflows (QC, clustering, marker identification)
- Proficient with analysis platforms: Microreact, Pathogenwatch, Data-flo
- Biological database retrieval & submission: ENA, NCBI, SRA, GEO, etc.

Workflow Development & Scripting

- Languages: Python, R, Shell, Nextflow (DSL2), CWL
- Pipeline reproducibility and deployment: Docker, Conda
- Automation and reproducible reporting using Jupyter

PUBLICATIONS (ORCID: [0000-0003-2775-2280](https://orcid.org/0000-0003-2775-2280))

1. Nagaraj G, **Shamanna V**, et al. Tracking the evolution of emerging serotypes and antibiotic resistance patterns in *Streptococcus pneumoniae* among Indian adults using high-throughput genome sequencing. *BMC Infect Dis.* 2025 Sep 29;25(1):1181. doi: [10.1186/s12879-025-11587-x](https://doi.org/10.1186/s12879-025-11587-x).
2. Bhatia M, **Shamanna V**, et al. Genomic characterization and antimicrobial resistance profile of *Salmonella* spp. clinical isolates: A single-centre hospital-based exploratory study from Uttarakhand, India. *Microb Pathog.* 2025 Aug;205:107686. doi: [10.1016/j.micpath.2025.107686](https://doi.org/10.1016/j.micpath.2025.107686).
3. Danielle J. Ingle, ..., **Varun Shamanna**, et al. Typhi Mykrobe: fast and accurate lineage identification and antimicrobial resistance genotyping directly from sequence reads for the typhoid fever agent *Salmonella* Typhi (2024). *bioRxiv* 2024.04.05.587953; doi: <https://doi.org/10.1101/2024.09.30.613582>
4. Kara K. Tsang, ..., **Varun Shamanna**, et al. Diversity, functional classification and genotyping of SHV β -lactamases in *Klebsiella pneumoniae* (2024). *Microb Genom.* 2024;10(10):001294.; doi: <https://doi.org/10.1099/mgen.0.001294>.
5. **Varun Shamanna**, et al. "Geographical distribution, disease association and diversity of *Klebsiella pneumoniae* K/L and O antigens in India: roadmap for vaccine development." *Microbial genomics* vol. 10,7 (2024): doi:[10.1099/mgen.0.001271](https://doi.org/10.1099/mgen.0.001271)
6. Bhatia, M., **Varun Shamanna**, et al. Assessment of in vitro colistin susceptibility of Carbapenem-resistant clinical Gram-negative bacterial isolates using four commercially available systems & Whole-genome sequencing: A diagnostic accuracy study. *Diagn. Microbiol. Infect. Dis.* 116155 (2023) doi:[10.1016/j.diagmicrobio.2023.116155](https://doi.org/10.1016/j.diagmicrobio.2023.116155).
7. Bhatia M, **Varun Shamanna**, Nagaraj G, et al. An insight into whole genome sequencing data of methicillin-resistant *Staphylococcus aureus* circulating in a teaching hospital in North India. *Indian J Med Microbiol.* 2023;44:100365. doi:[10.1016/j.ijmmb.2023.100365](https://doi.org/10.1016/j.ijmmb.2023.100365)
8. Monica I. Abrudan, **Varun Shamanna**, Akshatha Prasanna, et.al., Novel multidrug-resistant sublineages of *Staphylococcus aureus* clonal complex 22 discovered in India. *mSphere* 8, e0018523 (2023). DOI: <https://doi.org/10.1128/msphere.00185-23>
9. Megan E. Carey,, **Varun Shamanna**, et.al; Global diversity and antimicrobial resistance of typhoid fever pathogens: Insights from a meta-analysis of 13,000 *Salmonella* Typhi genomes. *eLife* 12:e85867, (2023). DOI: <https://doi.org/10.7554/eLife.85867>
10. Mohit Bhatia, **Varun Shamanna**, et.al Molecular characterization of carbapenem-resistant *Klebsiella pneumoniae* clinical isolates: preliminary experience from a tertiary care teaching hospital in the Himalayas, *Transactions of The Royal Society of Tropical Medicine and Hygiene*, (2022); trab189, <https://doi.org/10.1093/trstmh/trab189>.
11. Ayorinde O Afolayan, Johan Fabian Bernal, June M Gayeta, Melissa L Masim, **Varun Shamanna**, et.al., Overcoming Data Bottlenecks in Genomic Pathogen Surveillance, *Clinical Infectious Diseases*, Volume 73,

Statistical Analysis & Visualisation

- Statistical modelling and inference in R
- Differential expression, clustering, PCA, and regression analysis
- Scientific figure design and editing using Figma, ggplot2, and Cytoscape

Systems & IT Administration

- Linux server management, HPC workflows, Slurm
- Network administration, including Sophos Firewall

Scientific Communication

- Manuscript writing, presentations, technical documentation, and training for clinicians and researchers

TRAINING AND CERTIFICATION COURSES

1. GenEpi-BioTrain, ECDC, 2025: Empowering AMR Research through R: Analysis & Visualisation
2. ESCMID-funded Observership University Medical Center Groningen, Netherlands 2025: Real-time Application of bacterial WGS for hospital outbreaks and patient management ([report](#)).
3. GHRU CADA Digital Epidemiology: Digital Epidemiology data tools and workflows for Integrating epidemiological, laboratory, and genomic data – CGPS, Big Data Institute, Oxford.
4. Train the Trainer: Capacity building for genomic surveillance of AMR in low- and middle-income countries – Wellcome Genome Campus

Issue Supplement 4, 1 December 2021, Pages S267–S274,
<https://doi.org/10.1093/cid/ciab785>

12. Silvia Argimón, Geetha Nagaraj, **Varun Shamanna**, Dharmavaram Sravani, et.al., Circulation of Third-Generation Cephalosporin Resistant Salmonella Typhi in Mumbai, India, Clinical Infectious Diseases, (2021); ciab897, <https://doi.org/10.1093/cid/ciab897>
13. Geetha Nagaraj, **Varun Shamanna**, et.al., High-Resolution Genomic Profiling of Carbapenem-Resistant Klebsiella pneumoniae Isolates: A Multicentric Retrospective Indian Study, Clinical Infectious Diseases, Volume 73, Issue Supplement 4, 1 December 2021, Pages S300–S307, <https://doi.org/10.1093/cid/ciab767>

CONFERENCES

IMMEM XIV 2025

14th International Meeting on Microbial Epidemiological Markers, 17 – 20 September 2025, Porto, Portugal. Poster presentation on “The evolutionary trajectory of Klebsiella pneumoniae ST147 in India is defined by multiple introductions and capsular shifts”

ICID 2024

20th International Congress on Infectious Diseases (ICID), 3–6 December 2024, Cape Town, South Africa. **Oral presentation** on “Decoding diversity of Streptococcus pneumoniae isolated from Indian adults: Genomic Insights into Lineages, Population Dynamics, and Resistance Patterns”

KLEBS2024

Klebsiella Epidemiology and Biology Symposium 2024, 20–22 November 2024, Paris, France. Poster presentation on “Geographical distribution, disease association and diversity of Klebsiella pneumoniae K/L and O antigens in India: roadmap for vaccine development”

WSPID 2023

The 13th World Congress of the World Society for Pediatric Infectious Diseases (WSPID 2023), 14–17 November 2023 in Durban, South Africa. **Oral presentation** on “Prevalence Of Multidrug-Resistant *Klebsiella pneumoniae* In Pediatric Patients In India: A Whole-Genome Sequencing Study”

ISSAD 2023

The 3rd International Symposium on *Streptococcus agalactiae* Disease, 16 – 18 October 2023, in Rio de Janeiro, Brazil. **Oral presentation** on “Genomic characterisation of virulence factors & Pathogenic Islands of *Streptococcus agalactiae* isolated from healthy Indian women”

Bioinformatics Education Summit 2023

It was a global platform for discussion, development and dissemination of advances, guidance and best practices for training and education in bioinformatics as part of a global network.

ECCMID 2023

33rd European Congress of Clinical Microbiology and Infectious Diseases, 15th–18th April, Copenhagen, Denmark. **Oral Presentation** on “Multiclonal dissemination of multidrug-resistant high-risk clone ST147 *Klebsiella pneumoniae* carrying hybrid plasmids in India”.

ISID 2022

Poster presentation on “Whole Genome Sequencing Reveals Diverse Modes

Advanced Course.

5. Bioinformatics Tools And Methods For Antimicrobial Resistance - PHA4GE, JPIAMR, and CLIMB-BIG-DATA.
6. Bacterial Genomes: Antimicrobial Resistance in Bacterial Pathogens- Wellcome Connecting Science
7. COVID-19 data analysis workshop - ARTICnetwork and CLIMB-BIG-DATA
8. Bacterial Genomes I: From DNA To Protein Function Using Bioinformatics - Wellcome Connecting Science

TRAINING CONDUCTED

Next Generation Sequencing (NGS): Applications and Data Analysis

11–15 November 2024, for undergraduate students

Digital Epidemiology with Applications in Public Health

16 – 18 July 2024, Oxford, United Kingdom, ([link](#))

Linux Commands and Bash Scripting

For undergraduate students, 8–10 Jan 2024, Dayananda Sagar College of Engineering.

Using a genomic approach to characterise *Streptococcus pneumoniae* and *Streptococcus agalactiae*

12–18 Feb 2023, Central Research Laboratory, for GPS project partners from 12 countries.

Genomics for Clinicians

26–30 Sept 2022, Central Research Laboratory. For Clinicians across India from sentinel sites of the NIHR project.

Of Carbapenemase Gene Dissemination In Indian *Klebsiella pneumoniae* Clinical Isolates”

ISPPD-12

12th International Symposium on Pneumococci and Pneumococcal Diseases, Toronto, Canada, 19-23 June 2022.

Poster presentation on “Genomic Characterization Of Novel Lineages Of *Streptococcus pneumoniae* Reveals Dominance Of Non-Vaccine Serotypes In Indian Isolates”.

Antimicrobial Resistance – Genomes, Big Data and Emerging Technologies 2022

Poster presentation on “Large retrospective WGS study describes three major sequence types of *S. aureus* in India and reveals two novel multi-drug resistant sub-lineages of *S. aureus* Clonal Complex 22”

Antimicrobial Resistance – Genomes, Big Data and Emerging Technologies 2020

Poster presentation on “Population Structure, Antibiotic Resistance, and Pathogenicity of *Klebsiella pneumoniae*: A retrospective study from India.”

PROJECTS

In Central Research Laboratory:

Genomic Surveillance of Antimicrobial Resistance -2 Global Health Research Unit -India, funded by NIHR, UK - 2023-Present

Role: Digital Epidemiologist/Bioinformatician

Develop protocols & Pipelines for Nanopore Sequencing data analysis, including base calling, Assembly, Assembly Polishing, Variant Calling, and Plasmid Reconstruction.

Global Pneumococcal Sequencing (GPS) Project – India Node, funded by the Bill & Melinda Gates Foundation – 2023–Present

Role: Bioinformatician

Analyse pneumococcal genomes using the GPS pipeline for serotyping, GPSC typing, and AMR prediction, and contribute curated datasets to global surveillance. Lead CRL sub-study on surface proteins and non-typeable isolates for vaccine-relevant insights.

ANTIBACTERIAL ACTIVITY OF FAROPENEM AGAINST CLINICAL ISOLATES FROM INDIAN PAEDIATRIC INFECTIONS

2023-2024

Role: Project Coordinator

Assessing invitro activity of Faropenem using Estrips against Paediatric infections caused by *S. aureus*, *S. agalactiae*, *S. pneumoniae*, *K. pneumoniae*, and *S. typhi*. Comparing the results with other antibiotic classes and characterising genes contributing to Faropenem resistance.

Genomic Surveillance of Antimicrobial Resistance (Global Health Research Unit -India), funded by NIHR, UK - 2019-2023 March

Role: Data Analyst & Lab Infrastructure Development

Analysing pathogen sequence data, basic anonymised patient information,

Hands-on Workshop on Introduction to Bioinformatics

23-25 Aug 2022, NMAM Institute of Technology, Nitte. For Undergraduate students.

ACHIEVEMENTS

IMMEM XIV 2025

Best poster award

ICID 2024

Bill and Melinda Gates Foundation Travel Grant To attend the ICID 2024 conference held in Cape Town, South Africa

Digital Epidemiology with Applications in Public Health 2024

Invited faculty for the course conducted by CGPS & ESCMID on 16-18 July 2024 at Oxford, UK

WSPID 2023

Bill and Melinda Gates Foundation Travel Grant To attend the WSPID 2023 conference held in Durban, South Africa

Bioinformatics Education Summit 2023

Travel bursary from RITrainPlus - EBI, to attend the ‘Bioinformatics Education Summit 2023 held on 15-17 May, in Wellcome Genome Campus, Hinxton, UK

ECCMID 2023 Outreach Grant

Travel grant to attend the 33rd ECCMID held on 13- 18th 2023 in Copenhagen, Denmark

Advanced Bioinformatics Training 2023

Trainer and Organiser for the workshop developed by the GPS & JUNO projects, “Using a genomic approach to characterise *Streptococcus pneumoniae* and *Streptococcus agalactiae*”. Conducted at Central Research Laboratory, KIMS, India from 12th to 18th February 2023.

COG-TRAIN

and information about how samples were collected. Understanding the changing profile of antimicrobial resistance enables public health programs to respond locally, regionally, and ultimately nationally.

REFERENCES

Dr. Julio Diaz Caballero

Senior Bioinformatician,
Ellison Institute of Technology,
Winchester House, The Oxford Science Park,
Oxford, UK
Contact No: +44 7585 841588
Email: jdiaz@eit.org, Profile: [LinkedIN](#)

Dr K L Ravi Kumar,

Professor Emeritus, Microbiology
PI- GHRU, India Unit - Genomic AMR surveillance
CRL, Kempegowda Institute of Medical Sciences,
Banashankari 2nd Stage, Bangalore-70
Contact No: +91-9880229905
Email: klravikumar@gmail.com, Profile: [LinkedIN](#)

Dr. Natacha Couto

Bioinformatician,
Centre for Genomic Pathogen Surveillance,
Big Data Institute, Oxford, UK
Contact No: +351 963 543 633
Email: natachamongecouto@gmail.com, Profile: [LinkedIN](#)

Global Educator for the Future
Learn course titled “Making
sense of genomic data: COVID-19
web-based bioinformatics”

COG-TRAIN

Global Contributor for the Future
Learn course titled “The power of
genomics to understand
SARS-CoV-2”

British Society for
Antimicrobial
Chemotherapy (BSAC)
Member

ISID 2022

Bill & Melinda Gates Foundation
Professional Development Award
To attend the ISID 2022
conference held in Kuala Lumpur,
Malaysia.

ISPPD 12

Travel grant to attend the 12th
International Symposium on
Pneumococci & Pneumococcal
Disease (ISPPD-12), held on 19-23
June 2022 in Toronto, Canada.

LANGUAGES

English, Kannada, Telugu, Hindi

DECLARATION

I hereby declare that the
information furnished above is true
to my knowledge.